Perception & Design

PSY 410: Data Science for Psychology

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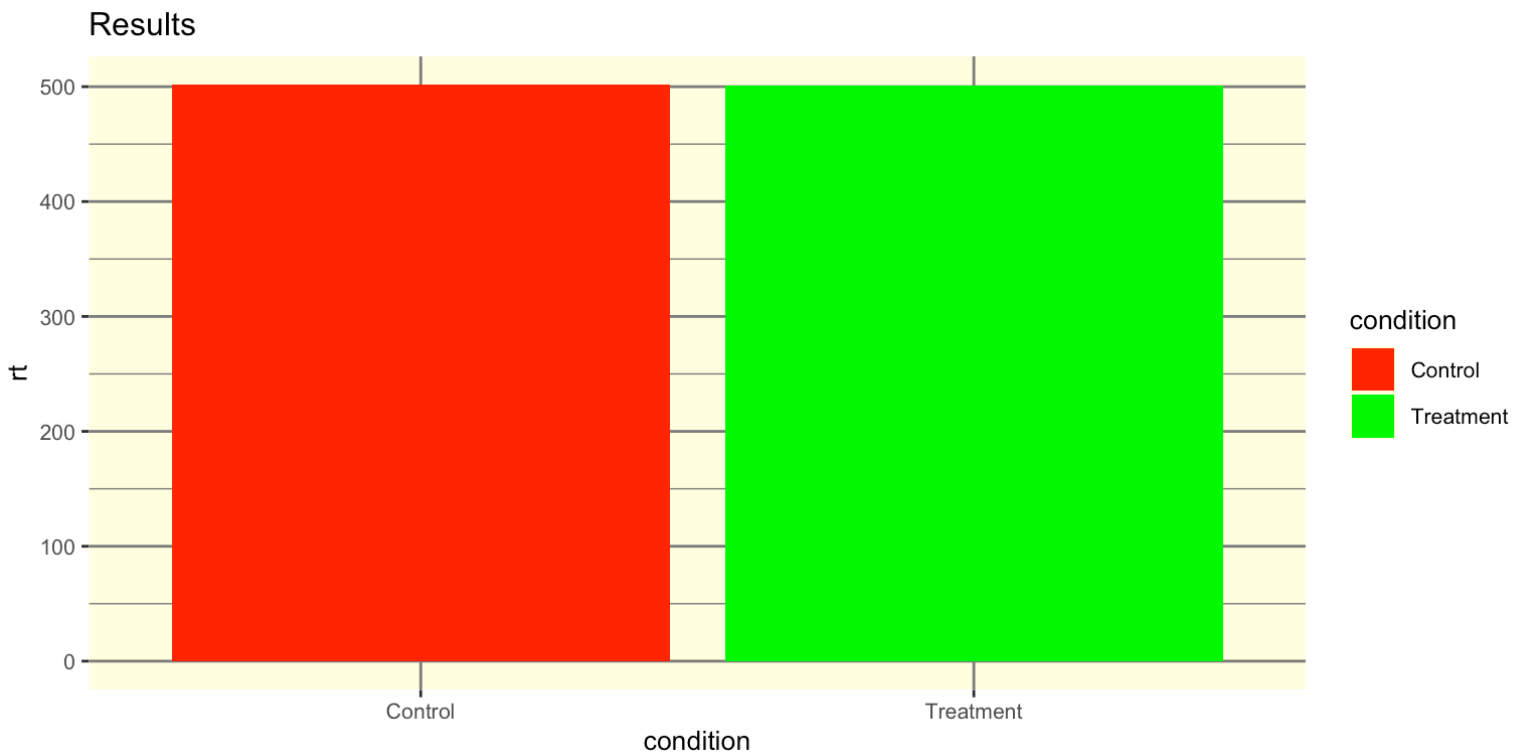
How we see data

What's wrong with this?



Red-green palette (colorblind-unfriendly). No informative title. Tiny text. Gridline noise. Bars hiding the data.

Today we learn **why your brain rejects bad figures** — and how to design ones that work.

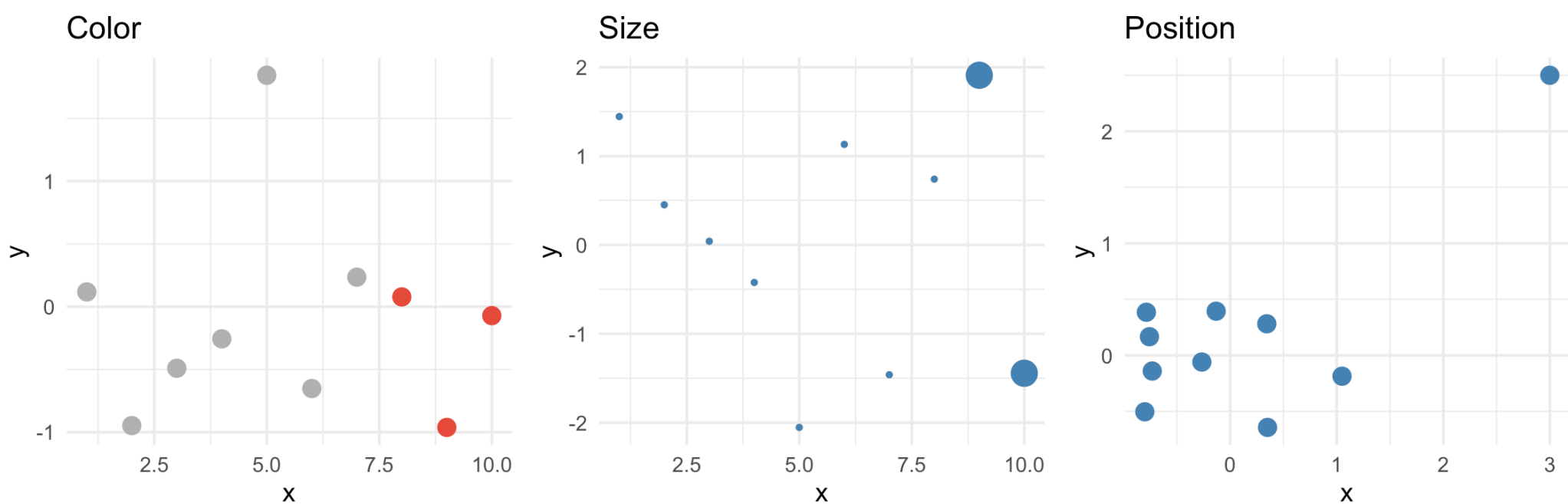
Your brain processes some visual features before you even think

Some visual features are processed almost instantly — before conscious attention kicks in:

Attribute	Example
Color	A red dot among blue dots
Size	A large circle among small ones
Position	A point far from the others
Shape	A triangle among circles
Orientation	A tilted bar among vertical bars

These are your tools for drawing the viewer's eye.

Preattentive in action



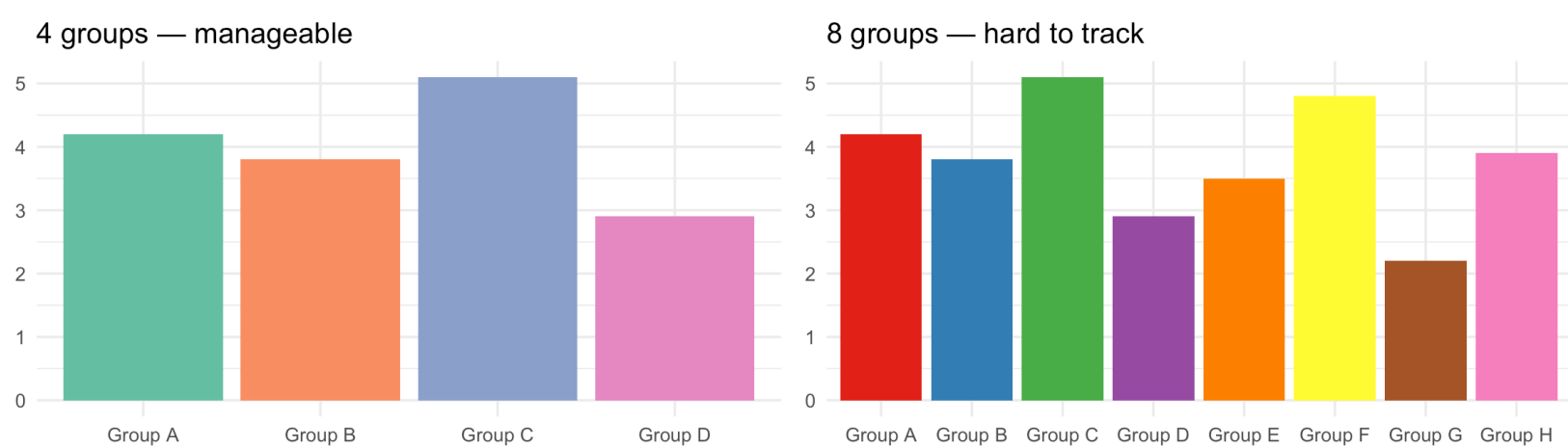
Your eye goes to the red points, the big points, and the outlier — instantly.

Working memory limits what you can encode

Research suggests working memory holds roughly **4 items** at a time (Cowan, 2001).

In data visualization, this means:

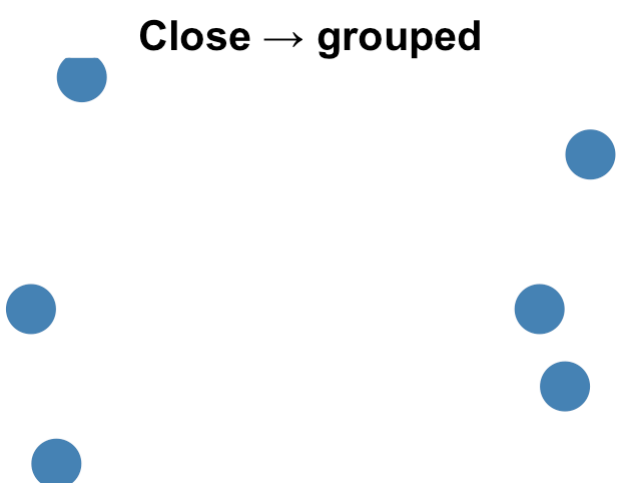
- Limit categorical groups in a legend to **4–5 maximum**
- Each additional aesthetic (color, shape, size) uses up a "slot"
- If you have 8 conditions, consider **faceting** rather than overlapping



Gestalt principles: how the brain groups things

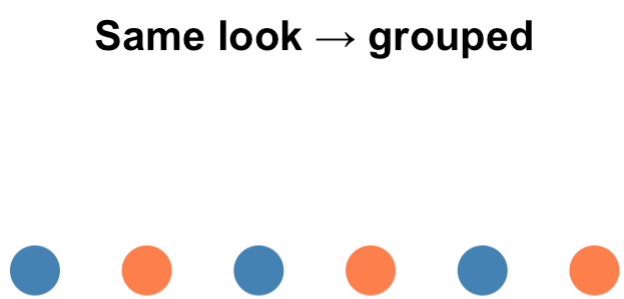
The brain automatically groups visual elements — whether you intend it or not.

Proximity



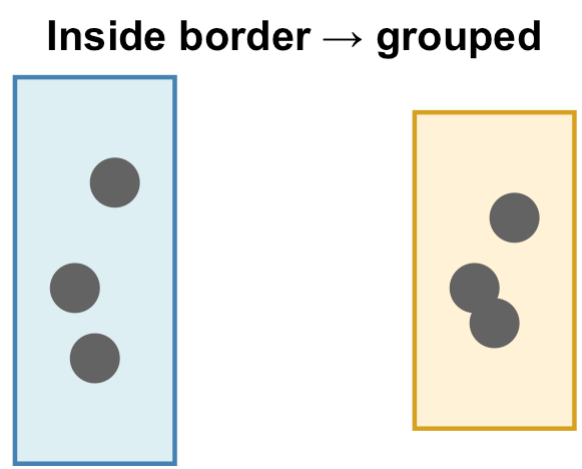
Nearby items feel like a group

Similarity



Same appearance = same group

Enclosure



Inside a border = same group

Gestalt principles show up everywhere in ggplot2

Principle	How it shows up in your figures
Proximity	`facet_wrap()` puts related panels side by side
Similarity	Color-coding makes same-group points feel connected
Enclosure	A shaded background or border sets a group apart
Continuity	Connecting points across time implies a single trend

These groupings happen whether you design them or not — use them intentionally.

Color theory

Three types of color palettes

Type	When to use	Example
Sequential	One continuous variable (low → high)	Blues, viridis
Diverging	Values relative to a midpoint	Red–white–blue
Qualitative	Categorical groups (no order)	Set1, tab10

Using the wrong type is one of the most common visualization mistakes.

First question: `fill` or `color`?

Which aesthetic did you map in `aes()`? That tells you which scale family to use.

```
# Geoms with `fill=`: bars, boxplots, tiles, density
ggplot(data, aes(x, y, fill = group)) + geom_boxplot() +
  scale_fill_manual(values = c(...))     # ← scale_fill_*

# Geoms with `color=`: points, lines
ggplot(data, aes(x, y, color = group)) + geom_point() +
  scale_color_manual(values = c(...))    # ← scale_color_*
```

Same pattern, just swap `fill` for `color` throughout.

Second question: which palette?

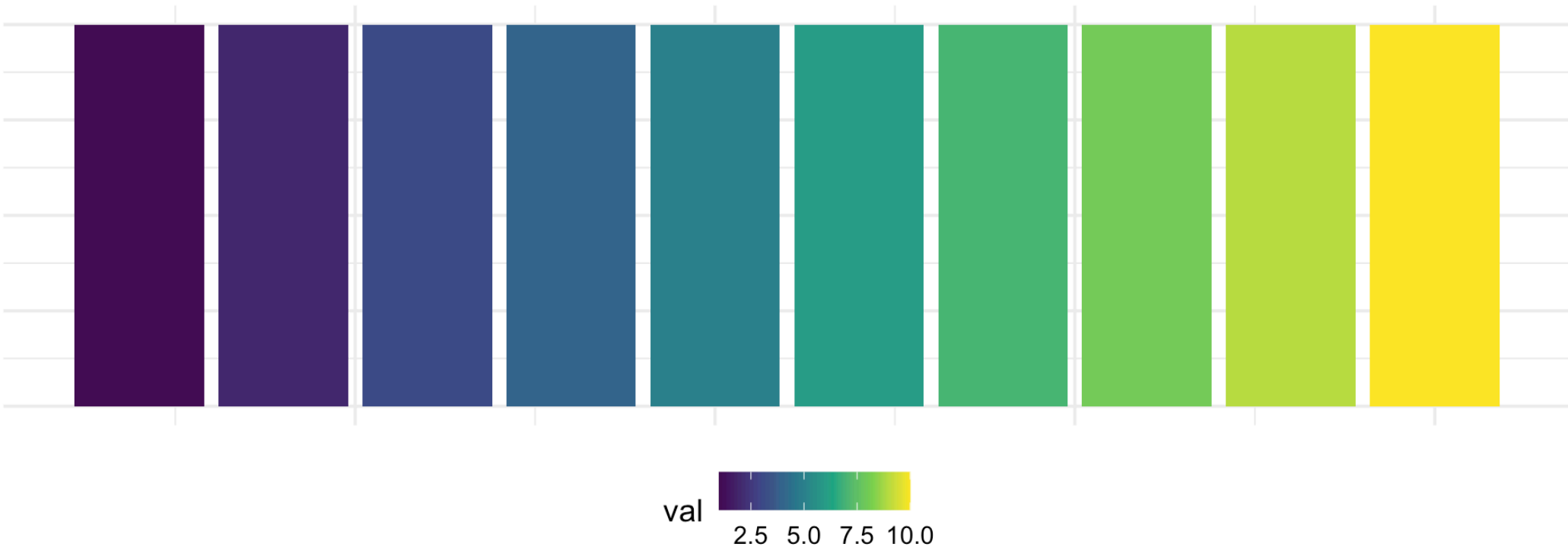
Match the palette to your data:

Your data	Function
Continuous numeric (e.g., reaction time)	`scale_fill_viridis_c()`
Ordered categories (e.g., dosage levels)	`scale_fill_viridis_d()`
Qualitative groups (e.g., therapy type)	`scale_fill_brewer(palette = "Set2")`
Diverging from a midpoint (e.g., change scores)	`scale_fill_distiller(palette = "RdBu")`
Hand-picked colors	`scale_fill_manual(values = c(...))`

`_c` for numeric, `_d` for character/factor. Same options work for `scale_color_*`.

Sequential palettes

viridis (sequential) — great for continuous data

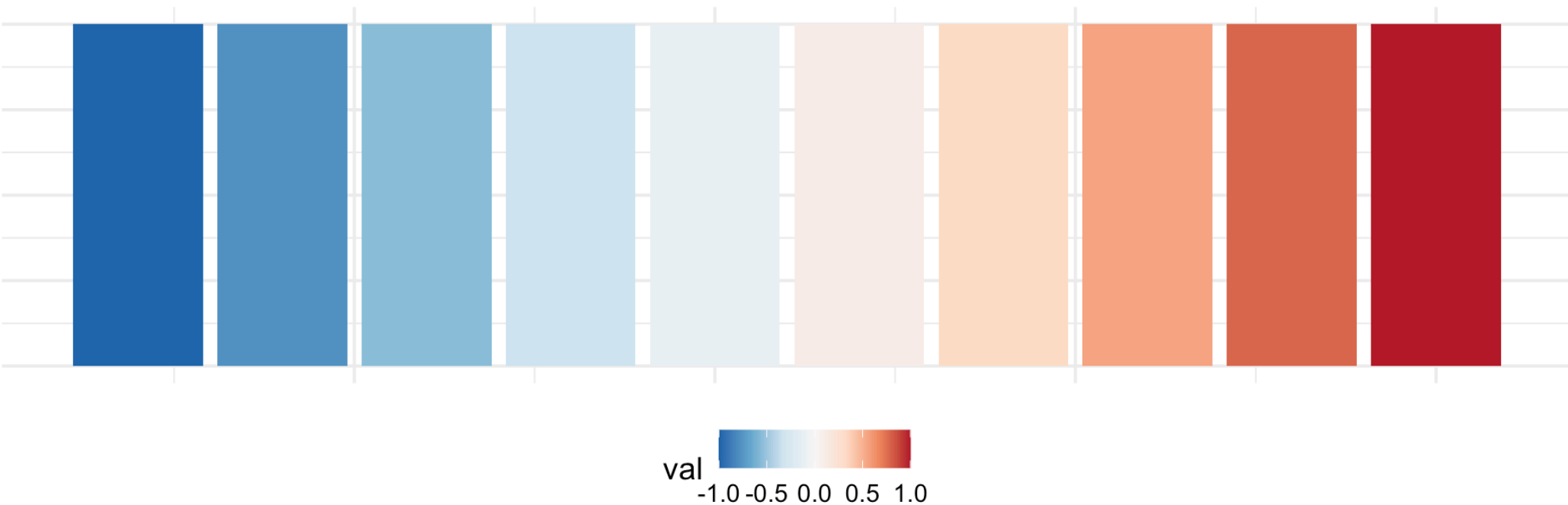


Diverging palettes

Use when values diverge from a meaningful **midpoint** — often 0 or a baseline.

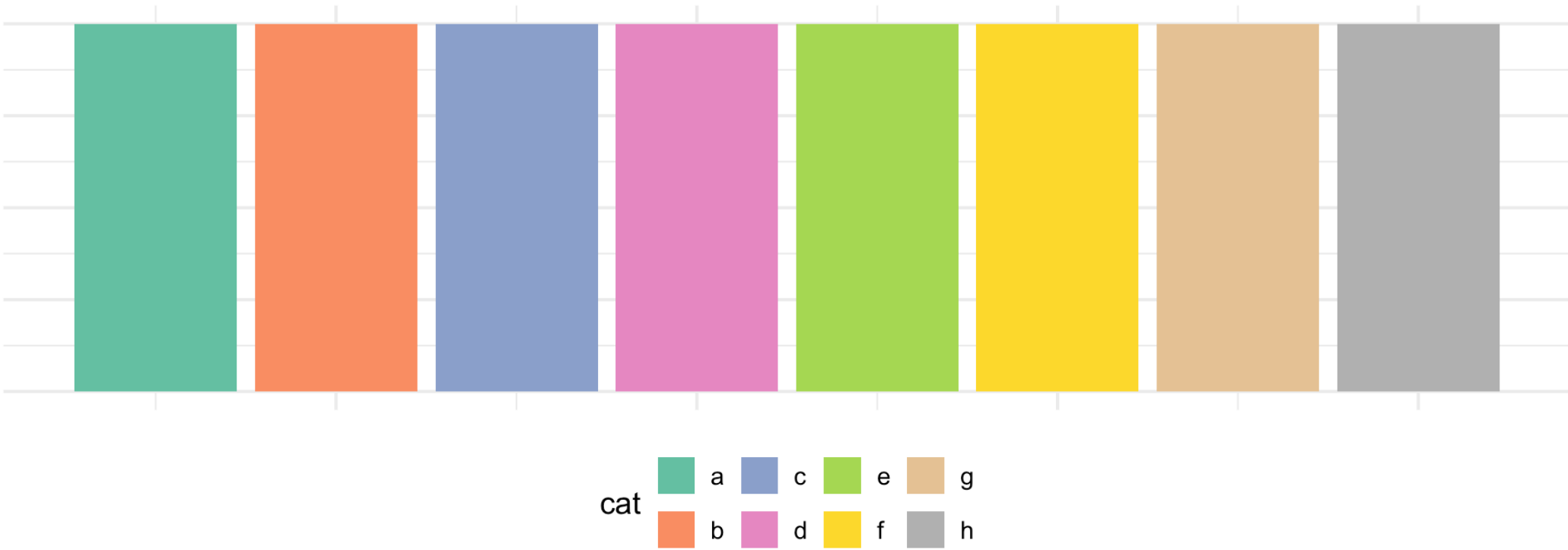
RdBu (diverging) — for data with a meaningful midpoint

Psychology examples: pre–post change scores, effect sizes, difference from baseline



Qualitative palettes

Set2 (qualitative) — for categorical groups



Colorblind-friendly choices

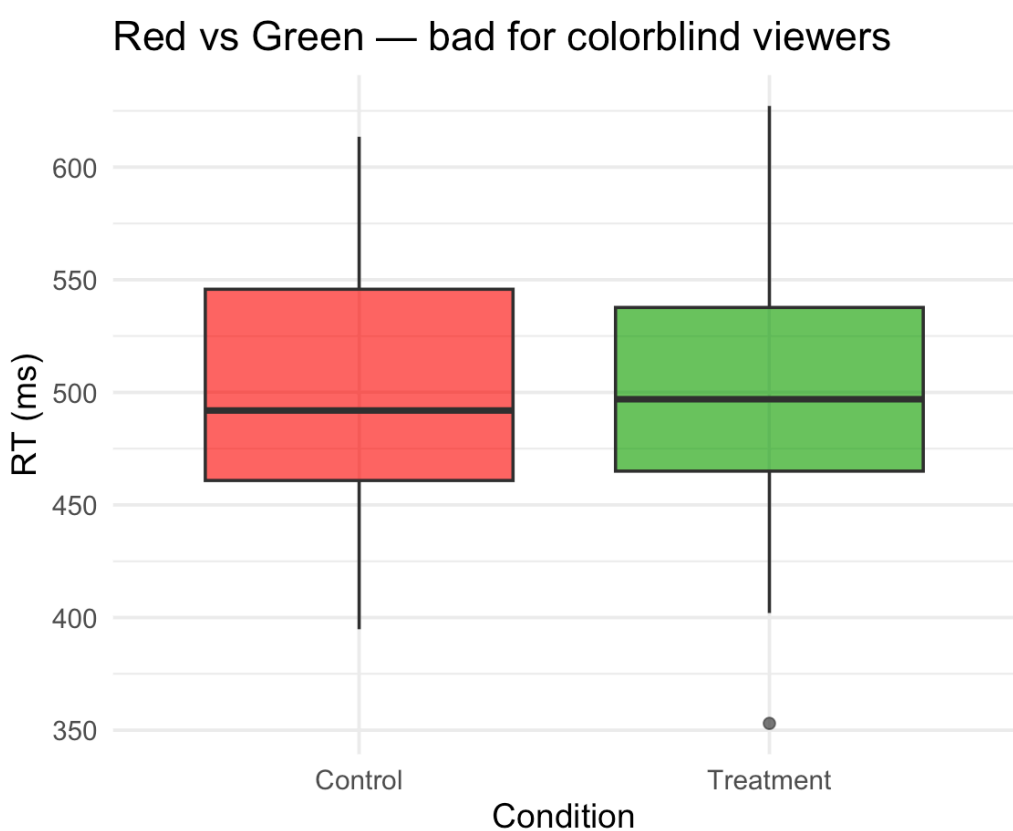
About 8% of men have some form of color vision deficiency. Red-green is the most common.

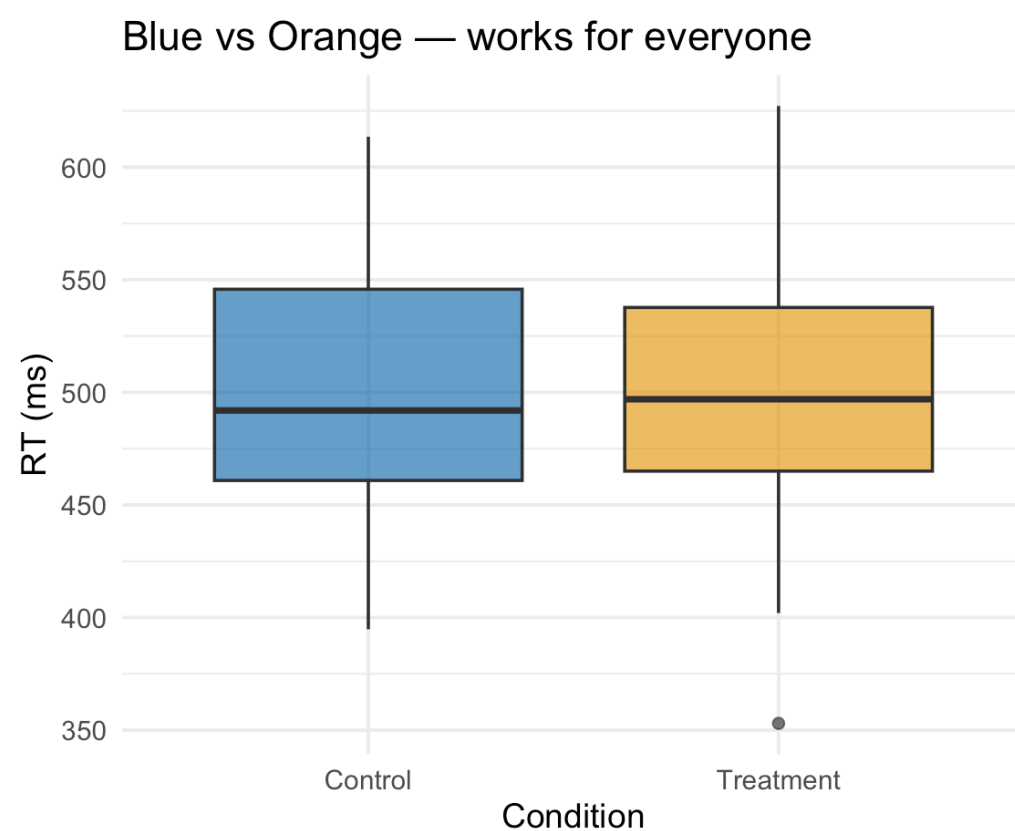
```
# viridis is colorblind-safe AND sequential
scale_fill_viridis_d()   # discrete
scale_fill_viridis_c()   # continuous

# ColorBrewer palettes designed for colorblindness
scale_fill_brewer(palette = "Set2")

# Or set colors manually with safe choices
scale_fill_manual(values = c("#0072B2", "#E69F00", "#009E73"))
# (blue, orange, green — distinguishable for most color vision types)
```

A bad color choice vs a good one





Why cluttered figures fail: cognitive load theory

Cognitive load theory (Sweller, 1988) says mental effort has three components:

Intrinsic load

The complexity of the data itself — you can't reduce it

Extraneous load

Clutter, noise, redundant elements — *this is what we control*

Germane load

The insight the reader builds — what we're trying to maximize

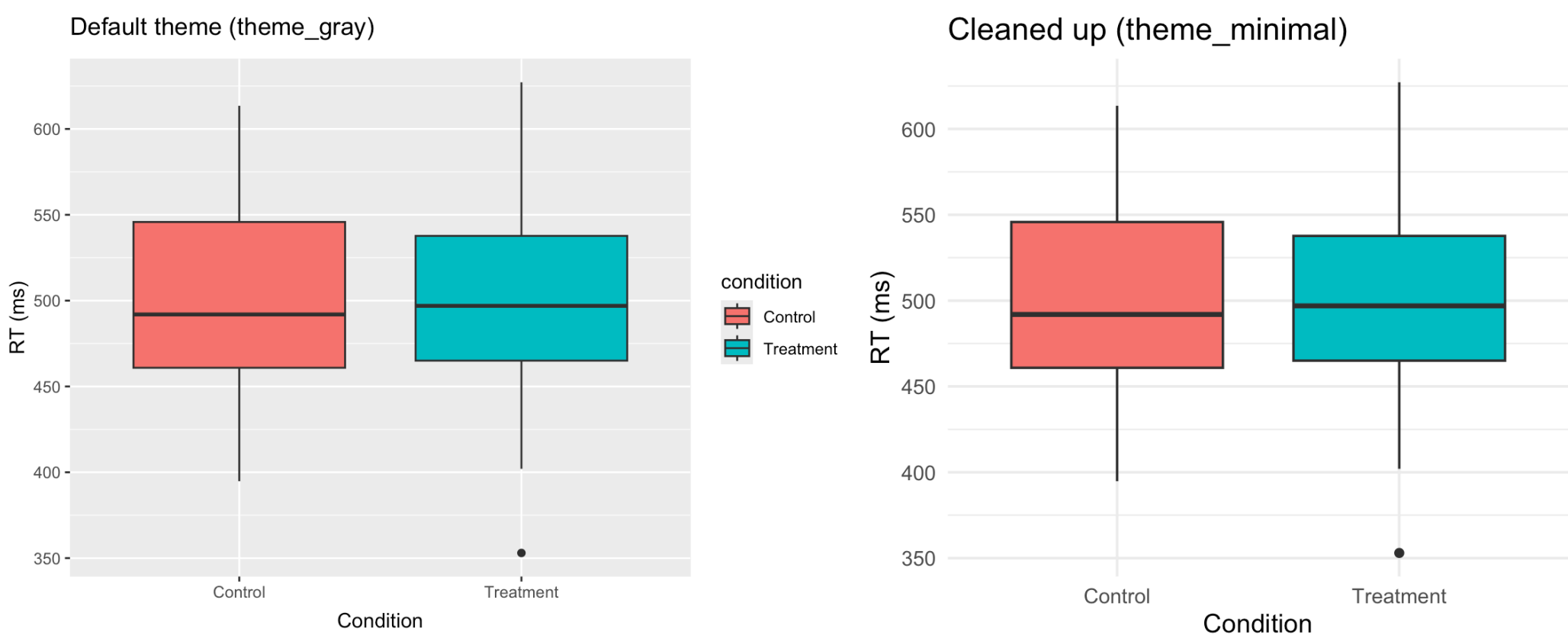
Every gray background, redundant legend, and unlabeled variable is **extraneous load** — it taxes working memory without adding information.

Design goal: minimize extraneous, maximize germane.

Decluttering

The default is cluttered

ggplot2's default theme adds a lot of visual noise. Compare:



The declutter checklist

Ask: does this element help the reader understand the data?

Remove

Element	Why
Gray background	Noise, no info
Gridlines (most)	Distraction
Redundant legend	X-axis says it
Generic axis labels	Add units instead

Keep

Element	Why
Title + subtitle	Orients the reader
Caption	Source, N, error bar type
Meaningful color	Highlights comparisons

Meet the `element_*` family

Every non-data piece of a ggplot is one of four kinds of thing:

Function	Key arguments	Used for
`element_text()`	`size`, `face`, `color`, `hjust`	Titles, axis text, legend text
`element_line()`	`color`, `linewidth`	Axis lines, gridlines, ticks
`element_rect()`	`fill`, `color`	Panel background, legend box
`element_blank()`	—	Removes the element entirely

`theme()` puts elements to use

Inside `theme()`, name the piece of the plot, then assign an `element_*` function to style it:

```
theme(
  plot.title      = element_text(face = "bold", size = 16),  # text → bold, l
  panel.grid      = element_line(color = "gray90"),           # lines → subtl
  plot.background = element_rect(fill = "white"),             # rect → white
  axis.ticks      = element_blank(),                          # blank → remov
  legend.position = "none"                                    # special: just
)
```

Not sure what an argument is called? Run `?theme` — there are ~90 options.

Pair coding break

Your turn: 10 minutes

I'll put a cluttered graph on screen. With a partner, **rewrite it** to follow the design principles we just covered.

Remove at least 5 unnecessary elements. Make it tell a clear story.

> **Tip**
>
> Think about: colors, legend, labels, theme, size mapping, and whether every aesthetic is adding information.

Your turn: 10 minutes

First, run this to create the dataset:

▶ Data setup (click to expand)

Now fix this:

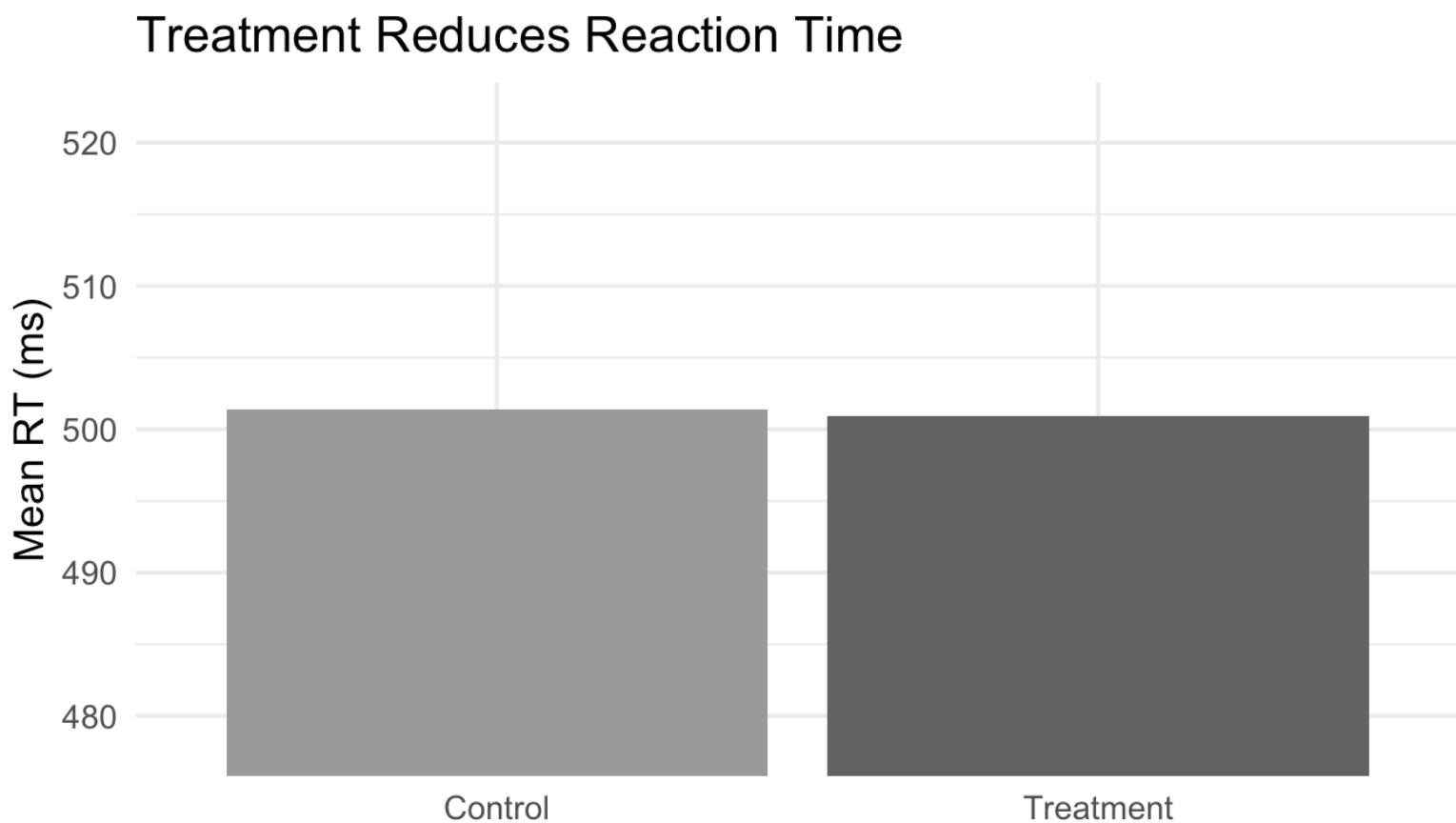
```
# The cluttered version — fix this!
reaction_data |>
  ggplot(aes(x = condition, y = rt, fill = condition, color = condition, size
  geom_point() +
  geom_boxplot(alpha = 0.3) +
  scale_fill_manual(values = c("Control" = "red", "Treatment" = "green")) +
  scale_color_manual(values = c("Control" = "red", "Treatment" = "green")) +
  labs(x = "condition", y = "rt", title = "data") +
  theme_gray()
```

Before we move on

📤 Submit your code on Canvas for participation credit. Paste what you have — it doesn't need to work perfectly.

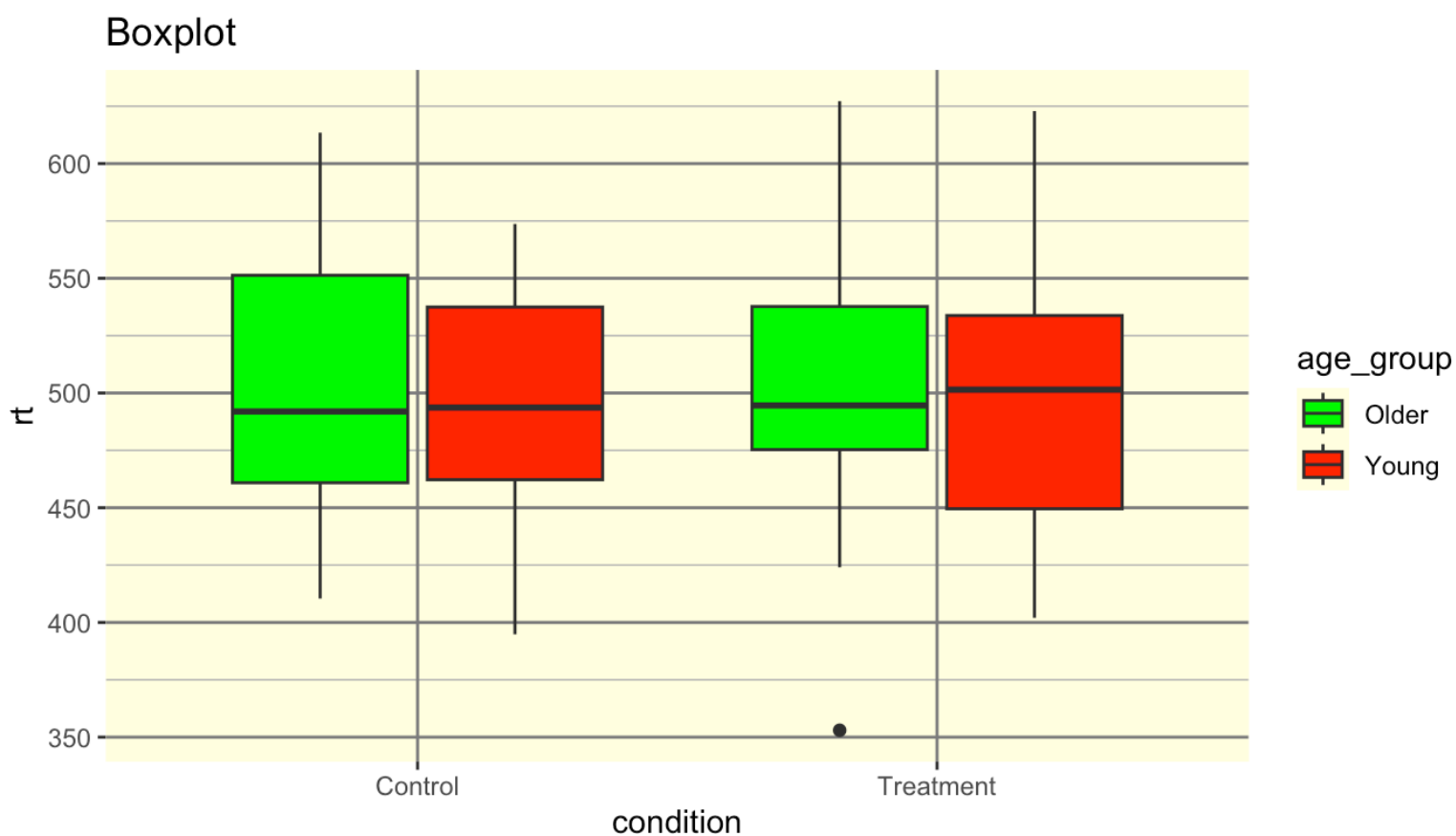
Critiquing bad graphs

What's wrong here? (1 of 3)



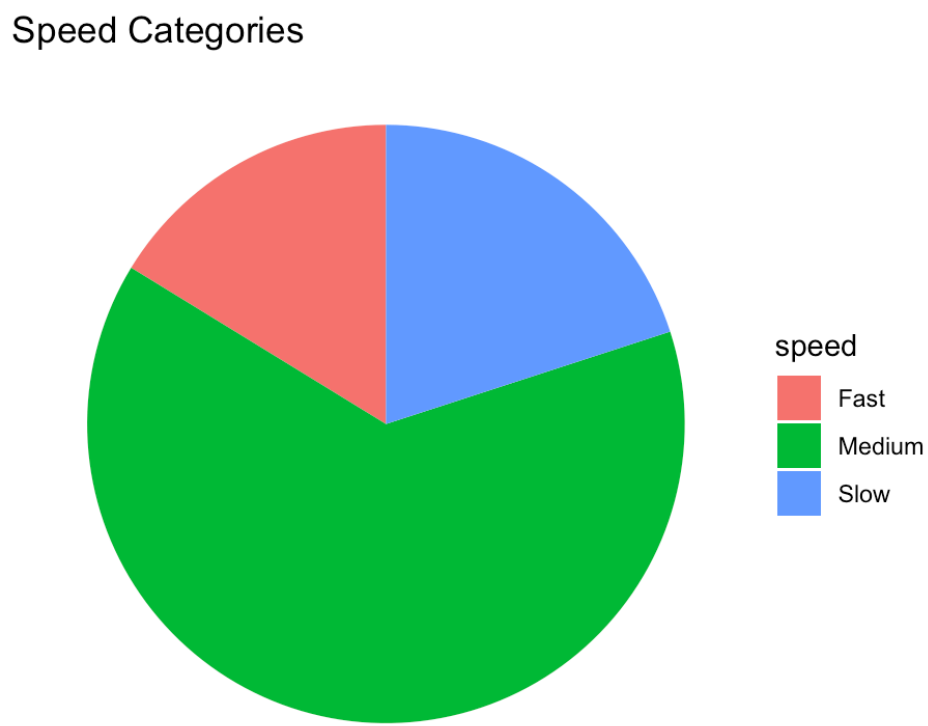
The problem: The y-axis starts at 478, not 0. The difference looks massive — but it's only ~40 ms. A truncated axis exaggerates the effect.

What's wrong here? (2 of 3)



The problems: Red-green palette (colorblind-unfriendly). Title says "Boxplot" (a label, not a finding). Variable names as labels. Distracting background color and gridlines.

What's wrong here? (3 of 3)



The problems: Continuous data was binned into arbitrary categories, then displayed as a pie chart. We lost the actual reaction times, the condition comparison, and the ability to see distributions. A histogram or density plot would show far more.

Get a head start

Assignment 4 preview

Assignment 4 will ask you to:

1. Create a **"bad" version** of a figure — deliberately violate design principles
2. Create a **"good" version** following what we covered today
3. Create a **colorblind-accessible version**

Assignment 4 preview

Start experimenting now:

- Take the `reaction_data` dataset
- Make the worst possible version of a figure
- Then make it great
- What did you change?

Wrapping up

Design principles checklist

- [ ] Is the geom appropriate for the data type?
- [ ] Are colors colorblind-friendly?
- [ ] Is the legend necessary? (Remove if redundant)
- [ ] Are axis labels informative (not just variable names)?
- [ ] Is there a title that states the main point?
- [ ] Is there visual clutter to remove?
- [ ] Do error bars have a caption explaining what they show?

Before next class

📖 **Read:**

- R4DS Ch 10: Exploratory data analysis (sections 10.1–10.4)

✅ **Practice:**

- Try the "bad vs good" exercise on your own
- Find a graph online and identify ways to improve it

Key takeaways

1. **Preattentive processing** — your brain detects color, size, and position before conscious thought
2. **Working memory** — limit categories to 4–5; every aesthetic uses a slot
3. **Gestalt principles** — proximity, similarity, and enclosure create grouping automatically
4. **Cognitive load** — minimize extraneous clutter so the insight can land
5. **Palette choice** — match sequential, diverging, or qualitative to your data type
6. **Design for colorblindness** — always; 1 in 12 men has color vision deficiency

These are not design opinions — they are perceptual psychology.

The one thing to remember

You're not designing for yourself. You're designing for a reader who will look at your figure for five seconds.

Next time: Exploratory Data Analysis